

An Acupuncture Robot Integrating Needle Bending Compensation and Manipulative Techniques: Modeling, Control, and Validation

Weihaio Li¹, Junlong Tao, Yu Han¹, Wanquan Liu¹, *Senior Member, IEEE*,
and Jianqing Peng², *Member, IEEE*

Abstract—In light of the shortage of acupuncturists and issues such as subjectivity leading to inconsistent efficacy and variability in accuracy and stability, an eight-degree-of-freedom (DOF) dexterous acupuncture robot system is developed in this article. An effective model predictive control (MPC) scheme, integrating needle bending compensation and manipulative techniques, is proposed to ensure the safety, accuracy, and stability of acupuncture manipulative trajectory tracking control. Based on acupuncture's working scenarios, a dexterous 2-DOF acupuncture mechanism is designed. Then, the kinematic and acupoint models are established. To enhance the controller's robustness, trajectory tracking constraints are included as soft constraints in the objective function before needle insertion, leading to the development of a trajectory-constrained MPC (TCMPC) model. This model not only improves the tracking accuracy and convergence speed but also ensures the stability of the solution. Additionally, a needle bending correction trajectory control method, considering needle deformation, is derived to perceive and correct the needle bending deformation under complex force environments. Furthermore, an adaptive impedance control method, integrating feedforward-feedback control (FFC), is proposed to simultaneously control force and track manipulative techniques under various contact environments. The uniform ultimate boundedness of the closed-loop system is verified using the Lyapunov theory. Finally, the effectiveness of the proposed methods and prototypes is validated through numerical simulations and practical experiments.

Index Terms—Acupuncture robot, adaptive impedance control, bending deformation, model predictive control (MPC), trajectory planning.

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Weihaio Li, Junlong Tao, and Wanquan Liu are with the School of Intelligent Systems Engineering, Shenzhen Campus of Sun Yat-sen University, Shenzhen 518107, China (e-mail: liwh77@mail2.sysu.edu.cn; taojlong@mail2.sysu.edu.cn; liuwq63@mail.sysu.edu.cn).

Yu Han and Jianqing Peng are with the School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen 518107, China, and also with Guangdong Provincial Key Laboratory of Fire Science and Technology, Guangzhou 510006, China (e-mail: hanyu25@mail.sysu.edu.cn; pengjq7@mail.sysu.edu.cn).

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I. INTRODUCTION

AS A SIGNIFICANT branch of traditional Chinese medicine (TCM), acupuncture is a unique and comprehensive therapy [1], [2] that has been proven effective in treating various diseases and pains [3], [4]. The World Health Organization reports that acupuncture is recognized in over 100 countries, with increasing international acceptance [5]. However, traditional acupuncture faces several challenges: 1) the lengthy training period for acupuncturists fails to meet the growing demand; 2) subjective factors cause variations in acupoint selection, depth, direction, and angle, leading to inconsistent results; and 3) physical and mental fatigue significantly reduces the accuracy and stability of acupuncturists.

Recent advances in intelligent robotic technology in medical engineering present a promising solution [6], [7]. Combining traditional acupuncture with intelligent robots can address issues of resource scarcity, subjectivity, and inconsistency. An acupuncture robot, typically comprising a manipulator, acupuncture mechanism, and visual sensors [8], remains largely in the theoretical or auxiliary medical stages. Xu and Xia [9] detailed the procedures for acupuncture robots. Lan and Litscher [10] focused on facial acupoint recognition. Du et al. [11] developed a virtual acupuncture system for training. Generally, an acupuncture robot operates in two steps: pre-insertion trajectory tracking based on visual positioning, followed by safe force/position control using accurate acupoint measurements. However, there is currently no relevant research on emulating TCM manipulative techniques for safe control in intelligent acupuncture robots.

Before acupuncture, the robot should move from its initial pose (i.e., position and orientation) to the target pre-insertion pose while considering a series of stringent environmental constraints [12]. In this context, the robot's movements are coupled with the reaction forces exerted on the environment [13], [14]. Kouro et al. [15] proposed a model predictive control (MPC) method that uses the receding horizon principle for optimal control by predicting the system's evolution. Typically, trajectory tracking control in the robot's Cartesian space is nonlinear, which leads to nonlinear MPC (NMPC) problems [16], [17]. The effectiveness of its solving methods, such as differential dynamic programming (DDP) [18] or linear-time varying (LTV)-MPC [19], depends on the system's nonlinear complexity. For multiple degree-of-freedom (DOF)

manipulators, these methods may exhibit poor performance. For nonlinear discrete systems, a simple but powerful control method is linear MPC. Limon et al. [20] proposed a linear MPC method for tracking piecewise constant references, ensuring offset-free tracking and convergence. Betti et al. [21] introduced a robust linear MPC method for handling bounded disturbances and infeasible references, demonstrating its noise resilience. Avanzini et al. [22] discussed a multiobjective constraint-based MPC for mobile robot tracking control. However, the trajectory between the robot and the target point, generated through trajectory optimization, remains uncertain and depends on sensor accuracy.

During the contact process, ensuring safe interaction between the acupuncture mechanism and human tissues through force and position control is a major technical challenge [23]. Given the flexibility of acupuncture needles in TCM and human tissues, controlling the deformation of the needle during contact is crucial. Most studies on needle deformation focus on beveled tip needles, attributing the bending to the unbalanced forces on the beveled tip. Roesthuis et al. [24] proposed a deformation estimation method based on needle insertion steps, deriving an energy model for needle deformation using the Rayleigh–Ritz method. Abayazid et al. [25] combined a needle deformation model with feedback from charge-coupled device images to achieve real-time steering control of flexible needles. Zhao et al. [26] introduced a preoperative needle insertion trajectory planning method, managing to model needle deflection, correct insertion angles, and optimize the needle trajectory. However, these studies primarily focus on trajectory pre-planning, with real-time control of deformation remaining an area for further research. Feng et al. [27] straightened the needle by linearly adjusting the robot's end position, but this method fails to consider orientation changes and cannot adapt to complex, nonflat interaction environments.

In acupuncture, the deformation–force relationship of the acupoint tissues is often modeled as a mass–spring system due to its simplicity, computational efficiency, and low resource requirements. Zeng et al. [28], [29] proposed a method utilizing this model for acupoint tissue deformation modeling under needle insertion. For soft environments susceptible to external disturbances, Wijayarathne et al. [30] introduced a framework for trajectory optimization and force control, equating contact forces to a Hunt–Crossley [31] model and integrating it with trajectory tracking using the alternating direction method of multipliers for coordinated force and position control. However, this approach requires offline model calibration and is not adaptable to varying contact environments. To address the uncertainty in the system model, Guo et al. [32] and Wang et al. [33] proposed a radial-basis-function neural network with adaptive updating weights, and Zhang and Khamesee [34] utilized an adaptive impedance adjustment method for environmental parameters. Xu et al. [35], [36] proposed a model-free impedance control method integrating position and force control, transforming the optimization problem into a recurrent neural network using Karush–Kuhn–Tucker conditions. While these methods address hybrid force–position control, they do not fully explore

synchronized motion and force control in one direction, limiting their use in acupuncture, where precise control of both force and position is critical.

To address the aforementioned problem, this article presents an intelligent acupuncture robot system that integrates needle bending compensation and TCM manipulative techniques. The system enables explicit operational space trajectory planning for obstacle avoidance and utilizes MPC for accurate, efficient, and robust trajectory tracking. It also controls the force and needle deformation during insertion to ensure safe and accurate contact operations. For high-fidelity planning, a beam-based force model [37] is employed to perceive needle bending and orientation, followed by trajectory planning and tracking for correction of any bending. Additionally, an adaptive impedance control method that accounts for the spring–damper contact model is proposed, allowing force control in various contact environments. It also facilitates real-time detection of contact force to autonomously switch between force control and manipulative technique tracking tasks.

In summary, the main contribution of this article is given as follows.

- 1) An 8-DOF acupuncture robot system is designed. It achieves safe and precise acupuncture operations.
- 2) A trajectory-constrained MPC (TCMPC) method is proposed. It features joint space formulation as a linear MPC and uniquely integrates a Cartesian trajectory tracking term in the objective function, enhancing trajectory correction and robustness. Compared to other advanced MPCs, the convergence speed and accuracy are improved by 42.48% and 44.08%, respectively.
- 3) A novel method for needle bending correction trajectory planning is introduced. It enables the detection and correction of needle bending in complex force environments and different needle lengths.
- 4) An adaptive impedance control method integrated feedforward-feedback control (FFC) is proposed. It achieves safe force control and manipulative technique tracking in complex acupuncture environments.

The remainder of this article is organized as follows. Section II presents the design of the acupuncture robot system and the tasks accomplished. Section III details the modeling of the acupuncture robot and describes the sensing process. Section IV introduces the trajectory planning control methods. Section V proposes the adaptive impedance control method for safe acupuncture. Section VI validates the effectiveness and accuracy of the model. Finally, Section VII concludes this article.

II. SYSTEM DESIGN

A. Design of the Dexterous Acupuncture Mechanism

The schematic of the acupuncture robot studied in this article is shown in Fig. 1(a). The patient lies on a treatment bed, and the system includes an 8-DOF acupuncture robot, two camera systems, a control host, and a patient model. The acupuncture robot consists of a 6-DOF manipulator and a 2-DOF dexterous acupuncture mechanism. The manipulator

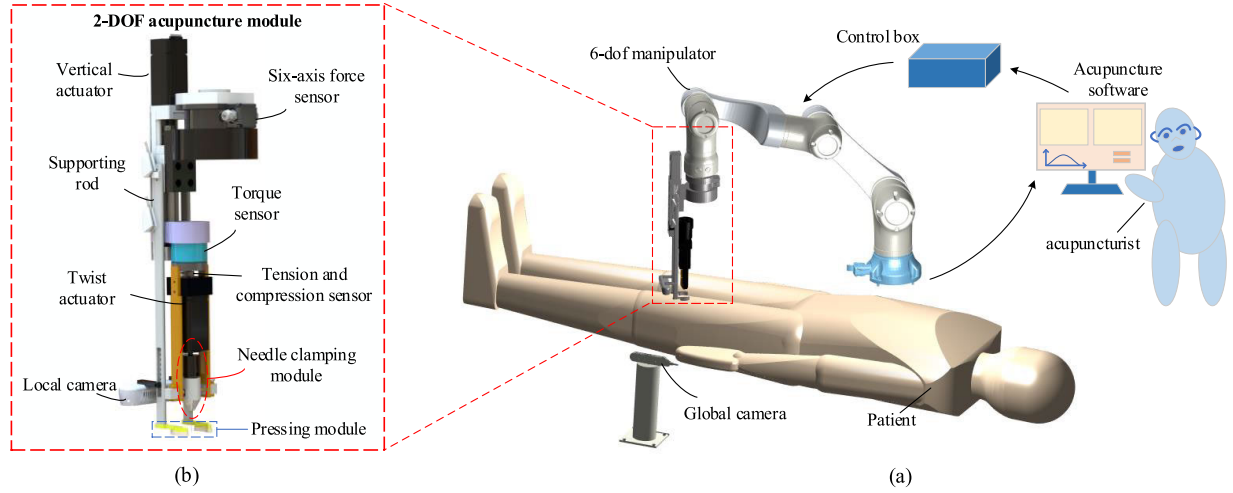


Fig. 1. Mechanism of the acupuncture robot. (a) Scene of the 8-DOF acupuncture robot surgery treatment. (b) Tips of 2-DOF acupuncture module.

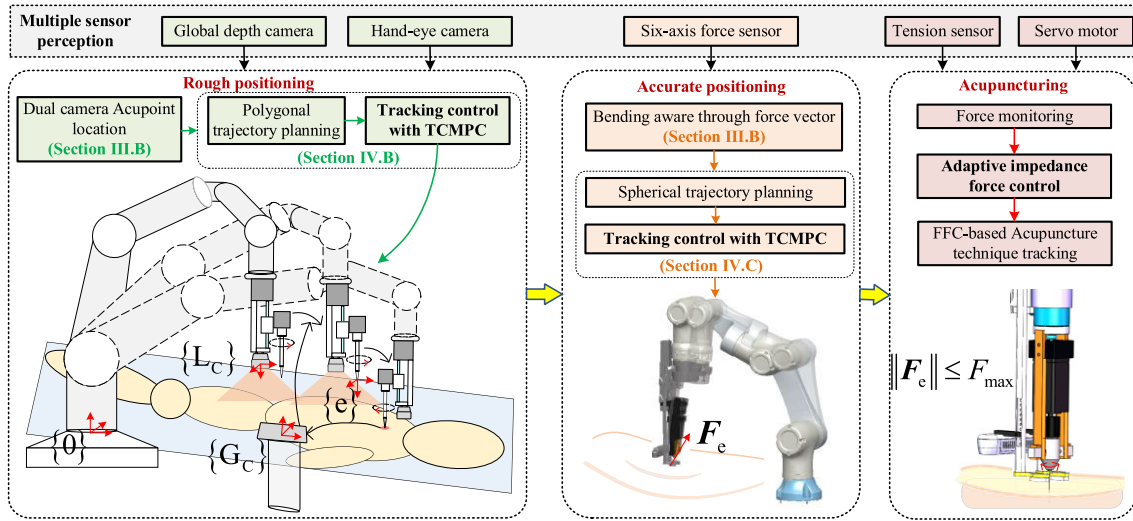


Fig. 2. Research framework of the acupuncture robot.

and acupuncture mechanism work together to precisely perform acupuncture procedures. The two camera systems include a global camera that assists in rough positioning and obstacle avoidance, and a local camera for hand-eye vision, mainly for accurate positioning of acupoints. The dexterous end-effector is a 2-DOF acupuncture mechanism that imitates TCM manipulative techniques, performing complex tasks such as lifting, inserting, and twisting. The 8-DOF acupuncture robot can implement the dexterous techniques of an acupuncturist.

The mechanical design schematic of the dexterous end-effector for needling is shown in Fig. 1(b). It adopts a standardized modular electromechanical system design [38] to ensure reliability at the sensing and mechanical levels. It includes a drive module, a sensor module, and an automatic needle-clamping module. The drive module consists of a lift-insert actuator that precisely controls the depth and speed of needle insertion and lifting, and a rotation actuator controls the needle's twisting movements. They are used to replicate the lift-insert and twisting techniques employed by acupuncturists. The force sensing module is equipped with a six-axis force/torque sensor at the end of the manipulator, as well as a one-axis torque sensor and a one-axis tension-compression sensor in the acupuncture mechanism. These sensors provide

detailed and accurate force feedback. The tension-compression sensor ensures that the needle can be inserted and lifted safely and accurately, while the torque sensor ensures precise and safe twisting of the needle. The automatic needle-clamping module is a crucial component of the dexterous acupuncture mechanism, facilitating needle installation and removal. The pressing device primarily stabilizes the skin tissue during the needling process.

B. Construction of the Acupuncture Robot System

The acupuncture robot system is constructed as shown in Fig. 2. The global camera and hand-eye camera collaborate to locate acupoint markers, providing precise pose information of the acupoints. Based on this information, the robot performs real-time trajectory planning and employs the TCMP model for trajectory tracking control during rough positioning. Upon reaching the target acupoint, the end-effector detects needle bending through a six-axis force/torque sensor. Real-time spherical trajectory planning is then carried out to correct the needle's bending. Precise trajectory tracking control is achieved using the TCMP model under accurate positioning. Throughout the needling process, observations from the tension-compression sensor are integrated with FFC

and adaptive impedance control, enabling safety tracking of acupuncture techniques. Consequently, the acupuncture robot system consists of a multisensor module, TCMPC models for both rough and accurate positioning, and an adaptive impedance control model. The automated system also follows the standard acupuncture disinfection procedures [39] based on the control methods.

III. MODELING AND LOCALIZATION OF THE ACUPUNCTURE ROBOT

A. Kinematic Model of the Acupuncture Robot

It is assumed that the joint angles of the n -DOF manipulator are $\mathbf{q} = [q_1, q_2, \dots, q_n]^T$; $\{0\} \sim \{n\}$ denotes the corresponding joint coordinate system and ${}^i T_j$ represents the homogeneous transformation matrix (HTM) of $\{j\}$ relative to $\{i\}$. According to the chain rule, the kinematic model of the acupuncture robot can be developed, that is,

$${}^0 T_n(q_1, q_2, \dots, q_n) = {}^0 T_1 {}^1 T_2 \dots {}^{n-1} T_n. \quad (1)$$

Let the Jacobian matrix from the joint space to the task space be $\mathbf{J}_n$. The 3-D position vector of the end-effector in $\{0\}$ is ${}^0 \mathbf{p}_n = [p_x, p_y, p_z]^T$, and the corresponding end-effector orientation represented in roll, pitch, and yaw (RPY) is ${}^0 \boldsymbol{\varphi}_n = [\alpha, \beta, \gamma]^T$. The velocity kinematics equation of the acupuncture robot can then be expressed as

$${}^0 \dot{\mathbf{X}}_n = \mathbf{J}_n(\mathbf{q}) \dot{\mathbf{q}} \quad (2)$$

where ${}^0 \dot{\mathbf{X}}_n = [{}^0 \dot{\mathbf{p}}_n^T, {}^0 \dot{\boldsymbol{\varphi}}_n^T]^T$, $\mathbf{J}_n(\mathbf{q}) = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_3 \end{bmatrix} \mathbf{J}$, $\mathbf{I}_3$ is the 3×3 identity matrix, $\mathbf{R}_3$ is the rotation matrix form of end-effector orientation, $\mathbf{J}$ is the geometric Jacobian matrix, and

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_v \\ \mathbf{J}_\omega \end{bmatrix} = \begin{bmatrix} \mathbf{J}_{v_1} & \mathbf{J}_{v_2} & \dots & \mathbf{J}_{v_n} \\ \mathbf{J}_{\omega_1} & \mathbf{J}_{\omega_2} & \dots & \mathbf{J}_{\omega_n} \end{bmatrix} \quad (3)$$

$$\mathbf{J}_{v_i}/\mathbf{J}_{\omega_i} = \begin{cases} \mathbf{z}_i \times (\mathbf{p}_e - \mathbf{p}_i)/\mathbf{z}_i, & \text{if joint } i \text{ is revolute} \\ \mathbf{z}_i/0, & \text{if joint } i \text{ is prismatic} \end{cases} \quad (4)$$

where $\mathbf{z}_i$ represents the unit vector of the rotation axis or motion direction of the i th joint and $\mathbf{p}_e - \mathbf{p}_i$ is the position vector of the end-effector $\mathbf{p}_e$ relative to the i th joint $\mathbf{p}_i$.

According to (2), the joint velocity of the acupuncture robot is given by

$$\dot{\mathbf{q}} = \mathbf{J}_n^\#(\mathbf{q}) \cdot {}^0 \dot{\mathbf{X}}_n \quad (5)$$

where $\mathbf{J}_n^\#$ is the pseudoinverse of $\mathbf{J}_n$.

B. Acupoint Perception of the Acupuncture Robot

It is assumed that $\{G_C\}$ and $\{L_C\}$ represent the global camera coordinate system and the end-effector hand-eye camera coordinate system, respectively. The HTM from $\{G_C\}$ to $\{0\}$ is denoted as ${}^0 T_{G_C}$. $\{L_C\}$ is fixed on the tip of the acupuncture mechanism. Consequently, the HTM from $\{L_C\}$ to $\{0\}$ ${}^0 T_{L_C}$ can be described as

$${}^0 T_{L_C}(q_1, q_2, \dots, q_n) = {}^0 T_1 {}^1 T_2 \dots {}^{n-1} T_n {}^n T_{L_C}. \quad (6)$$

Assume that the 3-D coordinates of the acupoint in $\{L_C\}$ and $\{G_C\}$ are ${}^{L_C} \mathbf{p}_{ap}$ and ${}^{G_C} \mathbf{p}_{ap}$, respectively. The first stage of the trajectory tracking process utilizes the global camera to guide the acupuncture robot to the region of interest, while the local camera refines the positioning accuracy as the acupuncture mechanism approaches the acupoint. Suppose that the end-effector's pose at time k is ${}^0 \mathbf{X}_n^k = [{}^0 \mathbf{p}_n^k, {}^0 \boldsymbol{\varphi}_n^k]^T$. To ensure the stability of visual positioning, the rough positioning planning process maintains $\gamma = -\pi$. When the hand-eye camera precisely measures ${}^{L_C} \mathbf{p}_{ap}$, the pose of the acupoint can be represented as

$${}^0 \mathbf{X}_k^d = \left[[{}^0 T_{L_C} \cdot {}^{L_C} \mathbf{p}_{ap}]^T, {}^0 \boldsymbol{\varphi}_n^k \right]^T. \quad (7)$$

When the acupoint image moves out of the field of view of the hand-eye camera, replace ${}^{L_C} \mathbf{p}_{ap}$ with the position result ${}^{G_C} \mathbf{p}_{ap}$ from the global camera in (7). By substituting ${}^0 \mathbf{X}_k^d$ into (5), the joint velocity $\dot{\mathbf{q}}_k^d$ of the acupuncture robot is obtained, and $\mathbf{q}_k^d$ is then determined through integration.

Define the coordinate frame at the acupuncture head as $\{E\}$, with $\mathbf{z}'$ being its unit vector along the z -axis. When the needle tip contacts human tissue, the six-axis force/torque sensor at the end of the acupuncture robot measures the force/torque, denoted as $[{}^E \mathbf{F}_s^T, {}^E \boldsymbol{\tau}_s^T]^T$, where ${}^E \mathbf{F}_s = [F_x, F_y, F_z]^T$ and ${}^E \boldsymbol{\tau}_s = [\tau_x, \tau_y, \tau_z]^T$. Due to the extremely small value of ${}^E \boldsymbol{\tau}_s$ and its significant drift error, it is neglected in this study. Let θ be the angle between ${}^E \mathbf{F}_s$ and $\mathbf{z}'$. Ideally, $F_x = F_y = 0$ when the needle is inserted perpendicularly. However, due to the complex structure of human tissue, displacement may occur, causing a deviation between ${}^E \mathbf{F}_s$ and $\mathbf{z}'$. According to the cantilever beam theory, the component $\mathbf{F}_{exy}$ of ${}^E \mathbf{F}_s$ in the xy plane causes the needle to bend, with the magnitude of $\tilde{\theta}$ indicating the extent of the bending, as depicted in Fig. 3(a). If this bending is not corrected during acupuncture, it will intensify as the needle penetrates deeper, potentially leading to treatment failure or even harm to the patient.

According to [37], $\tilde{\theta}$ is positively correlated with $\|\mathbf{F}_{exy}\|$, and $\|\mathbf{F}_{exy}\| = 0$ will result in $\tilde{\theta} = 0$. Since

$$\|\mathbf{F}_{exy}\| = \|{}^E \mathbf{F}_s\| \sin(\theta). \quad (8)$$

If $\theta \rightarrow 0$, then $\|\mathbf{F}_{exy}\| \rightarrow 0$, and if $\tilde{\theta} \rightarrow 0$, then the needle bending correction is realized by planning $\mathbf{z}'$ of $\{E\}$ to align with ${}^E \mathbf{F}_s$.

IV. PRE-NEEDLING TRAJECTORY TRACKING CONTROL

A. TCMPC Modeling

Assuming that the n -DOF acupuncture robot is controlled through joint velocity, the discrete joint space model of the acupuncture robot at time step k can be expressed as

$$\mathbf{q}_{k+1} = \mathbf{A}_C \mathbf{q}_k + \mathbf{B}_C \mathbf{u}_k \quad (9)$$

where $\mathbf{A}_C = \mathbf{I}_n \in \mathbb{R}^{n \times n}$ denotes the identity state matrix, $\mathbf{B}_C = T_s \mathbf{I}_n \in \mathbb{R}^{n \times n}$ is the control matrix, T_s represents the discrete time interval, and the control input $\mathbf{u} = \dot{\mathbf{q}} = d\mathbf{q}/dt$ signifies the joint velocity of the acupuncture robot.

Define

$$\begin{cases} \delta \mathbf{q}_k = \mathbf{q}_k - \mathbf{q}_{k-1} \\ \delta \mathbf{u}_k = \mathbf{u}_k - \mathbf{u}_{k-1}. \end{cases} \quad (10)$$

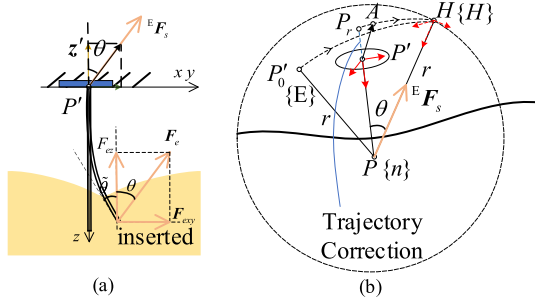


Fig. 3. Precision bending correction. (a) Force and deformation during needle insertion. (b) Spherical trajectory planning.

Set q_k^d as the target joint angle and $\hat{q}_k$ as the artificial reference joint angle. The tracking error between the actual joint angles and the reference value at time k is denoted by $e_k = q_k - \hat{q}_k$. The system's state vector is denoted by $\xi_k = [\delta q_k^T, e_k^T]^T$. Combining (9) and (10), the augmented system dynamic equation is expressed as

$$\xi_{k+1} = \underbrace{\begin{bmatrix} A_C & O_n \\ A_C & I_n \end{bmatrix}}_{A_v} \xi_k + \underbrace{\begin{bmatrix} B_C \\ B_C \end{bmatrix}}_{B_v} \delta u_k = A_v \xi_k + B_v \delta u_k \quad (11)$$

where O_n is a zero matrix of order n .

To improve the system's robustness against noise, a velocity-form MPC framework is adopted. Under the piecewise constant reference, the tracking problem can be expressed as

$$\begin{aligned} V_N^{O*}(\xi_k, q_k^d) &= \min_{\delta U_k, \hat{q}_k} L_v \\ &= \sum_{i=0}^{N-1} (\|\xi_{k+i}\|_Q^2 + \|\delta u_{k+i}\|_R^2) \\ &\quad + \|\xi_{k+N}\|_{P_{TC}}^2 + \|\hat{q}_k - q_k^d\|_V^2 \\ \text{s.t. vel-con} &= \begin{cases} \xi_{k+i+1} = A_v \xi_{k+i} + B_v \delta u_{k+i} \\ \delta u_{k+i} \in \mathbb{U}_\delta \\ \begin{bmatrix} q_{k+i+1} \\ u_{k+i} \end{bmatrix} \in \mathbb{Q} \times \mathbb{U} \\ \xi_{k+N} \in \mathbb{O}_\epsilon \end{cases} \end{aligned} \quad (12)$$

where $\delta U_k = [\delta u_k^T, \delta u_{k+1}^T, \dots, \delta u_{k+N-1}^T]^T$ is the increment control sequence defined over the prediction horizon N ; $Q \in \mathbb{R}^{2n \times 2n}$, $R \in \mathbb{R}^{2n \times 2n}$, $P_{TC} \in \mathbb{R}^{2n \times 2n}$, and $V \in \mathbb{R}^{2n \times 2n}$ are positive definite weight matrices; $\mathbb{Q} \times \mathbb{U}$ denotes the convex polygonal constraint set; $\xi_k = [\xi_k^T, \hat{q}_k^T]^T$; and $\mathbb{O}_\epsilon$ represents the maximal outputted admissible set.

For the trajectory tracking task of the acupuncture robot, q_k^d can be the steady reference for piecewise trajectories. However, it is difficult to strike a balance between the number of trajectory segments and the accuracy of trajectory tracking. The trajectory controlled by (12) is uncertain between set points and difficult to constrain. To address this problem, constraints for operational space tracking are added to (12), yielding

$$T_s J_i(q_i) u_{k+i} - S_{di} = 0 \quad (13)$$

where J_i represents the Jacobian matrix corresponding to the i th horizon and S_{di} denotes the target trajectory tracking value at that moment.

Equation (13) unifies the control step length and the tracking accuracy of the target trajectory, eliminating the uncertainty in trajectory tracking from (12). Considering the receding horizon principle of MPC, and to enhance the robustness of the controller, the constraints on trajectory tracking are added as soft constraints to the objective function. The TCMPC model is then proposed as

$$\begin{aligned} V_N^{FS*}(\xi_k, q_k^d) &= \min_{\delta U_k, \hat{q}_k} L_{Sp} = L_v + w \|T_s J_k u_k - S_d\|_2^2 \\ \text{s.t. vel-con} \end{aligned} \quad (14)$$

where S_d is the target trajectory tracking value for the first step of horizon and w represents the corresponding tracking weight.

The feasibility and stability of TCMPC are proven, as shown in Appendix A (see the Supplementary Material). The selection criteria of w are to some extent a tradeoff between the solvability and tracking accuracy. With a larger w , the tracking accuracy improves. However, when it reaches a certain value, the problem degenerates into LQR and the constraint is no longer soft. It is also demonstrated in Appendix A (see the Supplementary Material) and experimentally tuned in Section VI-A.

B. Trajectory Tracking Control Under Rough Positioning

For the working scenario of the acupuncture robot, the end-effector is planned to move along a right-angle trajectory to a safe position above the specified acupoint, as shown in Fig. 4. At the initial moment, the position of the end-effector is P_0 . The acupoint position measured by the vision system is O . The right-angle corner W can be determined by P_0 and O . Therefore, the trajectory of the end-effector during the rough planning stage is $P_0 \rightarrow W \rightarrow O$. Due to tracking errors, the actual end-effector position may deviate from the intended trajectory. Let P_h be the projection point of P on $\overrightarrow{P_0 W}$, and P_v be the projection point of P on $\overrightarrow{W O}$. The expected planning point at the next moment is A , and thus,

$$\overrightarrow{P_0 A} = P_h - P_0 + \lambda \cdot (W - P_h) \quad (15)$$

$$\overrightarrow{W A} = P_v - W + \lambda \cdot (O - P_v) \quad (16)$$

where λ represents the planning step size. In order to accelerate the convergence speed at the trajectory endpoints, λ is defined as

$$\lambda = \begin{cases} \frac{1}{\rho \cdot N}, & \|P - W\|_2 > \varepsilon_1 \text{ or } \|P - O\|_2 > \varepsilon_1 \\ \min\left(1, \frac{c}{\rho \cdot N}\right), & \|P - W\|_2 \leq \varepsilon_1 \text{ or } \|P - O\|_2 \leq \varepsilon_1 \end{cases} \quad (17)$$

where N is the MPC control horizon, ρ is a positive integer, c is the number of planning steps for segment $P_0 \rightarrow W$ or $W \rightarrow O$, and ε_1 is a small positive number.

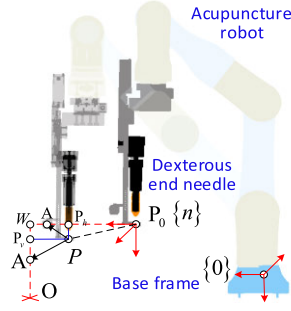


Fig. 4. Safe trajectory planning and tracking illustration.

In the next time step, the desired trajectory S_d planned for the end-effector of the acupuncture robot is

$$S_d = \overrightarrow{PA} = \begin{cases} \overrightarrow{P_0A} - \overrightarrow{P_0P}, & P \in P_0 \rightarrow W \\ \overrightarrow{WA} - \overrightarrow{WP}, & P \in W \rightarrow O. \end{cases} \quad (18)$$

When $\|P - W\|_2$ falls below ε_2 , which is smaller than ε_1 , the trajectory of $P_0 \rightarrow W$ is switched to $W \rightarrow O$. Therefore, the TCMPC model for rough positioning can be established as

$$\begin{aligned} \min_{\delta U_k, \bar{q}_k} L_{SR} &= L_v + w_r \cdot \|T_s J_{Tk} u_k - S_d\|^2 \\ \text{s.t.} \quad &\begin{cases} \text{vel-con} \\ J_{Rxy} u_k = 0 \\ q_n = q_{n-1} = 0 \end{cases} \end{aligned} \quad (19)$$

where J_{Tk} represents the transformation part of the Jacobian matrix, w_r denotes the tracking weight for rough positioning, and J_{Rxy} indicates the sub-Jacobian matrices corresponding to the roll and pitch degrees of freedom. The last constraint indicates that the acupuncture mechanism is unused in this stage.

The detailed trajectory tracking control method with rough positioning is shown in Appendix C.1 (see the Supplementary Material).

C. Trajectory Tracking Control Based on Bending Perception

Considering that needle bending may cause inaccuracy in the end-effector pose, the planning point is selected as the needle head point P' , and spherical planning is used to correct the needle's bending, as shown in Fig. 3(b). Assuming that the center of the sphere P is at the end-effector position without considering needle bending and the spherical radius r equals the needle's length, the expected point on the sphere is H . Consequently, the HTM for point P' is

$${}^0T_E = \begin{bmatrix} {}^0R_E & {}^0p_E \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \quad (20)$$

where 0R_E represents the rotation matrix from $\{0\}$ to $\{E\}$ and 0p_E denotes the 3-D coordinate of P' in $\{0\}$.

Then,

$$\overrightarrow{PH} = r \cdot \frac{{}^0R_E \cdot {}^E F_s}{\|{}^0R_E \cdot {}^E F_s\|}. \quad (21)$$

While

$$\overrightarrow{PP'} = {}^0p_n - {}^0p_E. \quad (22)$$

Thus,

$$\overrightarrow{P'H} = \overrightarrow{PH} - \overrightarrow{PP'} \quad (23)$$

$${}^0p_H = {}^0p_E + \overrightarrow{P'H} \quad (24)$$

where 0p_H denotes the 3-D coordinate of H in $\{0\}$.

It is obvious that θ is the angle between $\overrightarrow{PP'}$ and $\overrightarrow{PH}$. Let $k = [k_x, k_y, k_z]$ be their unit cross product vector. The rotation transformation matrix from $\{E\}$ to $\{H\}$ is given by

$${}^E R_H = I_3 + \sin \theta \cdot D + (1 - \cos \theta) D^2 \quad (25)$$

where

$$D = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}.$$

At this point, $\bar{q}_k^d$ can be written as

$$\bar{q}_k^d = J_n^\#(\bar{q}_{k-1}^d) \cdot \begin{bmatrix} {}^0\dot{p}_H^T & {}^0\dot{\phi}_H^T \end{bmatrix}^T \quad (26)$$

where ${}^0\phi_H = \text{Rot}_{f_{RPY}}({}^0R_E \cdot {}^E R_H)$, in which $\text{Rot}_{f_{RPY}}(\cdot)$ represents the function that converts the rotation matrix to RPY Euler angles.

The planned trajectory for this stage is arc. Due to tracking errors, P' will deviate from the spherical trajectory. The projection of P' onto the sphere is denoted as P'_r , and its 3-D coordinates can be expressed as

$${}^0p_{P'_r} = {}^0p_E + r \cdot \frac{\overrightarrow{PP'}}{\|\overrightarrow{PP'}\|}. \quad (27)$$

$\overrightarrow{PA}$ is obtained by rotating $\overrightarrow{PP'_r}$ around P in the $\lambda\theta$ angle, where λ in this stage is set to be

$$\lambda = \begin{cases} \frac{1}{\rho \cdot N}, & \|P - H\|_2 > \varepsilon_1 \\ \min\left(1, \frac{c}{\rho \cdot N}\right), & \|P - H\|_2 \leq \varepsilon_1. \end{cases} \quad (28)$$

According to (25) and (26), the corresponding rotation matrix is ${}^0R_{P'2A}$, and at this point

$${}^0p_A = {}^0R_{P'2A} \cdot {}^0p_{P'_r}. \quad (29)$$

Therefore, the desired planned trajectory for P' is

$$S_d = \overrightarrow{P'A} = {}^0p_A - {}^0p_E. \quad (30)$$

Incorporating the tracking error of S_d as a soft constraint into the objective function, the TCMPC model for this accurate positioning is given by

$$\begin{aligned} \min_{\delta U_k, \bar{q}_k} L_{AR} &= L_v + w_a \cdot \|T_s J'_{Tk} u_k - S_d\|^2 \\ \text{s.t.} \quad &\text{vel-con} \end{aligned} \quad (31)$$

where J'_{Tk} is the velocity Jacobian matrices corresponding to $\{E\}$ and w_a is the tracking weight in this stage.

The detailed trajectory tracking control method with accurate positioning is shown in Appendix C.2 (see the Supplementary Material).

Actually, the weight of the acupuncture mechanism affects the measurements of the six-axis force/torque sensor, which in turn influences the assessment of needle bending. For the acupuncture mechanism with gravity g , assuming that the sensor measurement value before denoising is ${}^E\hat{\mathbf{F}}_s$, the value ${}^E\mathbf{F}_s$ is compensated before being used for trajectory planning

$${}^E\mathbf{F}_s = {}^E\hat{\mathbf{F}}_s - ({}^0\mathbf{R}_E)^T \cdot \mathbf{v}_G \quad (32)$$

where $\mathbf{v}_G = [0, 0, -g]^T$.

V. ADAPTIVE IMPEDANCE CONTROL FOR MANIPULATIVE TECHNIQUE DURING NEEDLE INSERTION

When the robot reaches the desired pose, the acupuncture mechanism starts the needle insertion process. The manipulative technique can be quantified by the needle insertion speed $\dot{x}_d$ and depth x_d . A FFC is designed to regulate the speed commands sent to the motors, given by

$$u = \dot{x}_d + k_F(x_d - x_m) \quad (33)$$

where k_F is the proportional gain of the feedback and x_m is the measured needle insertion depth.

The force F_e exerted by the needle during acupuncture may exceed safety limits $F_{\max}$. Considering factors such as force mutation, tissue rebound, and safety during needling, an adaptive impedance control method combined with FFC is proposed. This approach achieves both technique tracking and needle force control in complex contact environments.

Typically, position-level impedance control can be represented as a mass-spring-damper system

$$M\Delta\ddot{x} + B\Delta\dot{x} + K\Delta x = \Delta F \quad (34)$$

where $\Delta F = F_r - F_e$ represents the error between the desired force and the measured force; $\Delta x = x_c - x_r$ represents the error between the actual position trajectory and the desired; and M , B , and K are the mass, damping, and stiffness coefficients, respectively.

Rewrite (34) to the transfer function

$$\Delta X(s) = G(s)\Delta F(s) = \frac{1}{Ms^2 + Bs + K}\Delta F(s) \quad (35)$$

which represents the model of needle-tissue force by a linear spring-damper model

$$F_e = \begin{cases} K_e(x_m - x_e) + B_e\dot{x}_m, & x_m \geq x_e \\ 0, & x_m < x_e \end{cases} \quad (36)$$

where K_e and B_e are the environmental stiffness and damping coefficients, respectively, and x_e is the environment position.

Then,

$$\Delta F = F_r - F_e = F_r - K_e(x_m - x_e) - B_e\dot{x}_m. \quad (37)$$

The trajectory x_c generated by the impedance filter is fed into the FFC controller. Ideally, the following holds:

$$\dot{x}_m = u = k_F(x_c - x_m) + \dot{x}_d. \quad (38)$$

Transforming (38) into the Laplace domain gives

$$s \cdot X_m(s) = k_F(X_c(s) - X_m(s)) + \dot{X}_d(s). \quad (39)$$

Combining (32), (34), and (36), and rearranging give

$$\begin{aligned} & \left(1 + \frac{K_e k_F}{s + k_F}G(s) + \frac{B_e s k_F}{s + k_F}G(s)\right)\Delta F(s) \\ &= F_r(s) + K_e X_e(s) - \frac{K_e}{s + k_F}(k_F X_r(s) + \dot{X}_d(s)) \\ & \quad - \frac{B_e s}{s + k_F}(k_F X_r(s) + \dot{X}_d(s)). \end{aligned} \quad (40)$$

Therefore, the steady-state error of the needling force can be expressed as

$$\Delta F_{ss} = \lim_{s \rightarrow 0} s \cdot \Delta F(s) = \frac{K \cdot K_e}{K + K_e} \left(\frac{F_r}{K_e} + x_e - x_r - \frac{\dot{x}_d}{k_F} \right). \quad (41)$$

From (41), define x_r as

$$x_r = \frac{F_r}{K_e} + x_e - \frac{\dot{x}_d}{k_F}. \quad (42)$$

Combining (41) with (42), $\Delta F_{ss} \rightarrow 0$, and the system can thereby control the operational force.

In fact, when there is a deviation between K_e , B_e , and x_e and their actual values, the steady-state error determined by (42) will be unpredictably affected. The spring-damper model varies among different individuals and tissues. Additionally, factors such as needle insertion depth and skin rebound also affect the accuracy of the spring-damper model. Therefore, it is necessary to estimate the model parameters in real time. Following the adaptive adjustment scheme for environmental parameters based on the Lyapunov function in [34], the final impedance control equation combined with FFC can be expressed as:

$$x_r = \frac{F_r}{\hat{K}_e} + \hat{x}_e - \frac{\dot{x}_d}{k_F} \quad (43)$$

$$\hat{K}_e(t) = \hat{K}_e(0) - \gamma_1 \int_0^t x_m(\hat{F}_e - F_e)dt \quad (44)$$

$$\hat{x}_e(t) = \hat{x}_e(0) + \gamma_1 \int_0^t \frac{(\hat{F}_e - F_e)}{\hat{K}_e} (x_m \hat{x}_e + \gamma_2/\gamma_1)dt \quad (45)$$

$$\hat{B}_e(t) = \hat{B}_e(0) - \gamma_3 \int_0^t \dot{x}_m(\hat{F}_e - F_e)dt \quad (46)$$

$$\hat{F}_e = \hat{K}_e(x_m - \hat{x}_e) + \hat{B}_e\dot{x}_m \quad (47)$$

where γ_1 – γ_3 are positive scalars, and $\hat{K}_e$, $\hat{B}_e$, and $\hat{x}_e$ are the predicted values of the spring-damper model parameters.

The control block of the acupuncture mechanism is illustrated in Fig. 5. To coordinate the tracking of the manipulative technique with force control, a switch module is adopted to autonomously adjust the input for the FFC position trajectory. When the acupuncture operation force does not exceed the force limit, i.e., $F_e - F_{\max} < \nu$, where ν is a positive scalar, the FFC position trajectory input is set to the desired trajectory x_d . When the force exceeds the limit, i.e., $F_e - F_{\max} \geq \nu$, $F_{\max}$ is regarded as F_r , and x_d is replaced with the outputted of the adaptive impedance filter x_c .

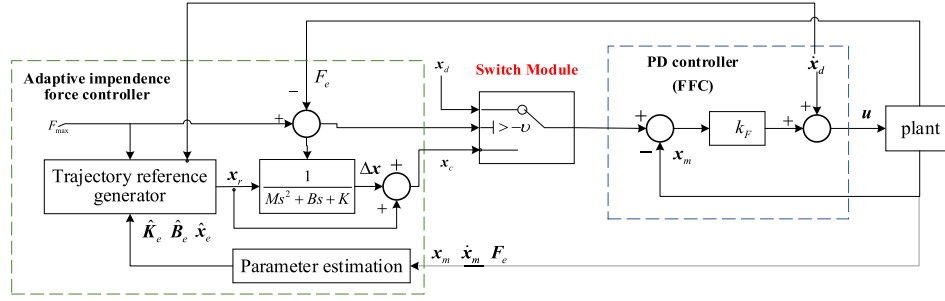


Fig. 5. Control block of the acupuncture mechanism.

VI. SIMULATION AND EXPERIMENT

A. Simulation

1) *System Setup*: In order to validate the proposed acupuncture robot modeling and control model, a simulation system is developed in MATLAB/Simulink. The 8-DOF acupuncture robot is modeled using the Robotics Toolbox, which includes a 6-DOF industrial manipulator and a 2-DOF needling module. The DH parameters are listed in Appendix D.1 (see the Supplementary Material). The simulation system runs on a Windows 10 computer with a configuration of 32-GB RAM and Intel i7-10700 CPU, 2.90 GHz. The TCMPC is solved online using qpOASES [40].

2) *Simulation Results*: They are given as follows.

a) *Trajectory tracking control simulation under rough positioning*: It is given as follows.

Case 1-1: In order to validate the capability of trajectory tracking under rough positioning, three initial points within the workspace are randomly selected. The end-effector orientations have an angle of $-\pi$ with the z -axis, and the selected initial points are (30, 266, 60), (212, -383, 102), and (45, 97, 67) mm. The corresponding initial joint angles in radians are $(-4.522, 0.279, 1.674, -0.140, 1.791, -2.635, 0, 0)$, $(-1.263, -1.259, -0.678, -2.940, -2.543, 4.735, 0, 0)$, and $(0.314, -1.394, -1.968, 2.9790, 2.5205, -1.5637, 0, 0)$. The target point is set to (500, 110, -100). The horizon N is set to 10; the weighting coefficients for trajectory tracking are $w_r = 5e2$, $\rho = 4$ with step size $\lambda = 1/(4N)$, $\varepsilon_1 = 5$ mm, and $\varepsilon_2 = 1$ mm. The tracking results for the polygonal trajectory are shown in Fig. 6(a), and the position tracking error and orientation variation curves are shown in Fig. 6(b).

Case 1-2: To further evaluate the tracking capability with different target points, the initial point is set at (45, 97, 67) with initial joint angles (0.314, -1.394, -1.968, 2.9790, 2.5205, -1.5637, 0, 0). The target points are randomly selected as: 1 (500, 110, -100), 2 (300, 300, -50), and 3 (400, -250, -200), all within the workspace. The trajectory for target point 4 (580, 100, -150) is out of the boundaries of the workspace. The TCMPC parameters are the same as before. The tracking results are shown in Fig. 6(c), and the tracking error and orientation variation are in Fig. 6(d).

Case 1-3: The initial point is set at (45, 97, 67) with initial joint angles of (0.314, -1.394, -1.968, 2.979, 2.5205, -1.5637, 0, 0). The target point is (500, 110, -100). In order to explore the effect of the tracking weight coefficients on tracking results, w_r is set to 5e1, 5e2, and 5e3 sequentially.

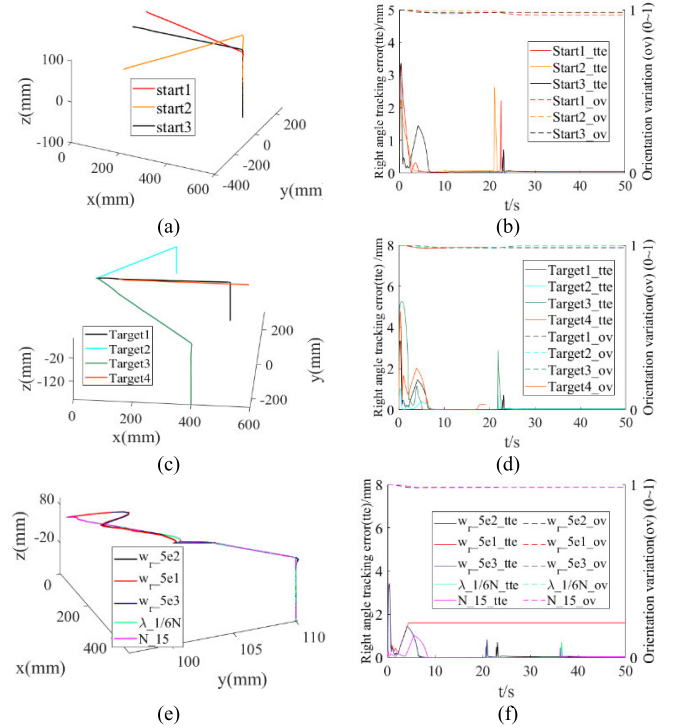


Fig. 6. Tracking performance. (a), (c), and (e) Polygonal trajectory planning and tracking results of cases 1-1, 1-2, and 1-3, respectively. (b), (d), and (f) Corresponding tracking error curves.

Then, fix $w_r = 5e2$, set $\rho = 6$ with step size $\lambda = 1/6N$, and set N to 15. The tracking results are shown in Fig. 6(e) and (f).

b) *Trajectory tracking control simulation based on bending perception*: It is given as follows.

Case 2-1: To verify the tracking capability of spherical trajectories, the initial joint angle of (1.634, 0.012, -1.01, 0.058, -2.15, 0.153, 0, 0) is selected within the workspace. ${}^E F_s$ is set to $(-1, -1, -2)$, (0.3, -1.5, -2), and (0.3, 1, -2) sequentially. $[{}^0 p_H, {}^0 R_H]$ is calculated by (20)~(24). Assume that $r = 30$ mm and the sphere center coordinate is (38.370, -297.800, 144.600). Set N to 10, $\rho = 3$ with step size $\lambda = 1/(3N)$, $\varepsilon_1 = 0.05$ mm, and trajectory tracking weight coefficient $w_a = 5$. The spherical tracking results are shown in Fig. 7(a), with corresponding position and orientation error curves in Fig. 7(b).

Case 2-2: With ${}^E F_s$ set as (0.3, -1.5, -2) and other settings unchanged, the tracking results for different needle lengths are shown in Fig. 7(c) and (d).

Case 2-3: Furthermore, with ${}^E F_s$ set as (0.3, -1.5, -2), the tracking results for w_a of 5, 5e1, and 5e3 and $w_a = 5$ and $N = 20$ are shown in Fig. 7(e) and (f).

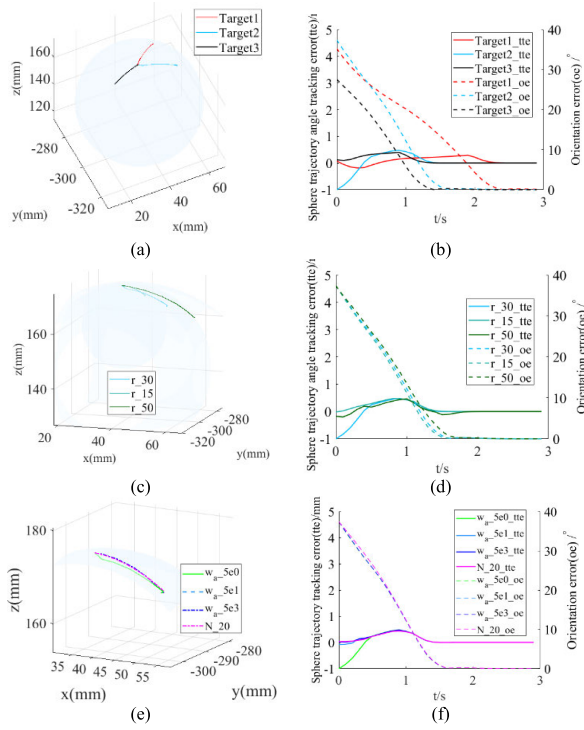


Fig. 7. Tracking performance. (a), (c), and (e) Sphere trajectory planning and tracking results of cases 2-1, 2-2, and 2-3, respectively. (b), (d), and (f) Corresponding tracking error curves.

c) *Adaptive impedance control simulation of manipulative techniques*: The simulation step size is 0.01 s, and the impedance filter in the continuous s -domain is

$$G(s) = \frac{1}{10s^2 + 500s + 600}. \quad (48)$$

Set the sampling time of the impedance filter to 0.05 s, and convert it to the discrete z -domain expression using the Tustin transformation as

$$G(z) = \frac{2.732e - 5z^2 + 5.464e - 5z + 2.932e - 5}{z^2 - 0.8415z - 0.0929}. \quad (49)$$

Let $k_F = 1$, $\gamma_1 = 5$, $\gamma_2 = 10$, and $\gamma_3 = 0.5$. The initial values of the predictive environment parameters are $\hat{K}_e = 100$, $\hat{B}_e = 0.5$, and $\hat{x}_e = 0$. The actual contact force model of the environment is

$$F_e = \begin{cases} 1.4x_m + 0.1\dot{x}_m, & x_m \geq 0 \\ 0, & x_m < 0. \end{cases} \quad (50)$$

Based on the lifting and inserting techniques in TCM, the designed velocity tracking model based on trigonometric and polynomial functions is

$$\dot{x}_d = \begin{cases} 17 \cos(\pi \cdot (t\%1)), & 0 < t < 2 \\ 13 \cos(\pi \cdot (t\%1)), & 2 < t < 4 \\ 10 \cos(\pi \cdot (t\%1)), & 4 < t < 10 \end{cases} \quad (51)$$

$$\dot{x}_d = \begin{cases} 0.006 \cdot t(t-3)(t-5)(t-8)(t-10), & 0 < t < 3 \\ 0.031 \cdot t(t-3)(t-5)(t-8)(t-10), & 3 < t < 5 \\ 0.0083 \cdot t(t-3)(t-5)(t-8)(t-10), & 5 < t < 8 \\ 0.0156 \cdot t(t-3)(t-5)(t-8)(t-10), & 8 < t < 10. \end{cases} \quad (52)$$

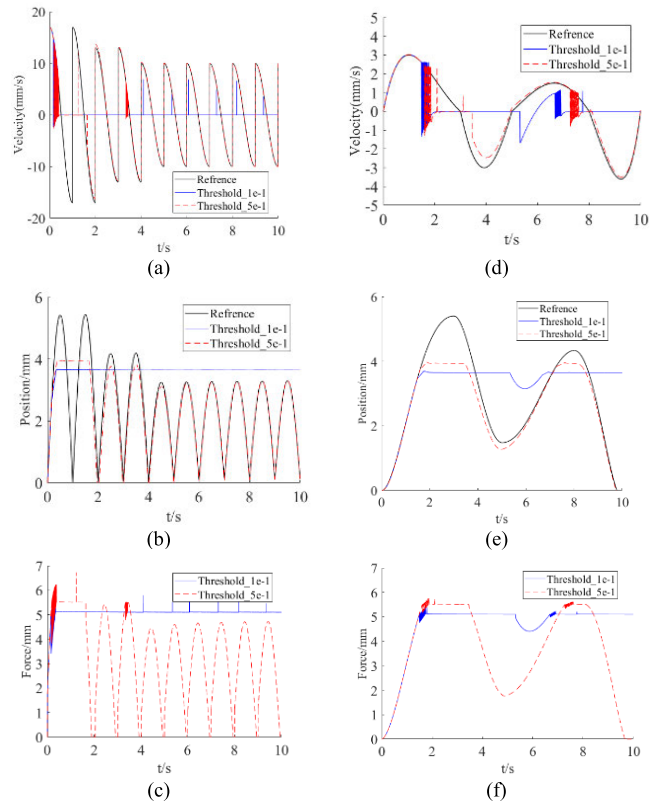


Fig. 8. Velocity tracking and force control performance of FFC + adaptive impedance controller. (a)–(c) Curves for velocity, position tracking, and force control result of the trigonometric acupuncture method, respectively. (d)–(f) Curves for velocity, position tracking, and force control result of the polynomial acupuncture method, respectively.

The sampling frequency of the switch module is set to match the simulation step size. Threshold ν is set to 0.1 and 0.5. Set $F_{\max} = 5\text{N}$. The tracking control for the techniques described by (51) and (52) is performed. The tracking results for velocity, displacement, and force control are shown in Fig. 8.

d) *Comparative simulation*: To highlight the advantages of the proposed TCMPC, two additional linear MPC methods are also used in this simulation verification for comparison, which are given by the following.

- 1) *Traditional MPC (TMPC)*: The formulation is exactly as described in (12).
- 2) *LTV-MPC [19]*: An online prediction controller based on a linear time-varying model, which is described as follows:

$$\begin{aligned} \min_{\delta \mathbf{U}_k} & \sum_{i=1}^{N_p} \|\mathbf{t}_i - \mathbf{t}_{\text{ref},i}\|_Q^2 + \sum_{i=0}^{N_c-1} \|\delta \mathbf{u}_i\|_{\mathbf{R}}^2 + \alpha \|\varepsilon\|_1 \\ \text{s.t.} & \begin{cases} \tilde{\mathbf{t}}_{i+1} = \begin{bmatrix} A_i & B_i \\ I & 0 \end{bmatrix} \tilde{\mathbf{t}}_i + \begin{bmatrix} B_i \\ I \end{bmatrix} \delta \mathbf{u}_i \\ \begin{bmatrix} \delta \mathbf{u}_i^T & \mathbf{q}_{i+1}^T & \mathbf{u}_i^T \end{bmatrix}^T \in \mathbb{U}_\delta \times \mathbb{Q} \times \mathbb{U} \\ \mathbf{S}_{di} - \varepsilon \leq T_s \mathbf{J}_i(q_i) \mathbf{u}_{k+i} \leq \mathbf{S}_{di} + \varepsilon \end{cases} \end{aligned} \quad (53)$$

where the tracking is based on the task space state $\mathbf{t}_i$ to track the reference $\mathbf{t}_{\text{ref},i}$, with

$$\mathbf{t}_{i+1} = \mathbf{t}_i + T_s \mathbf{J}_i(q_i) \mathbf{u}_i = A_i \mathbf{t}_i + B_i \mathbf{u}_i. \quad (54)$$

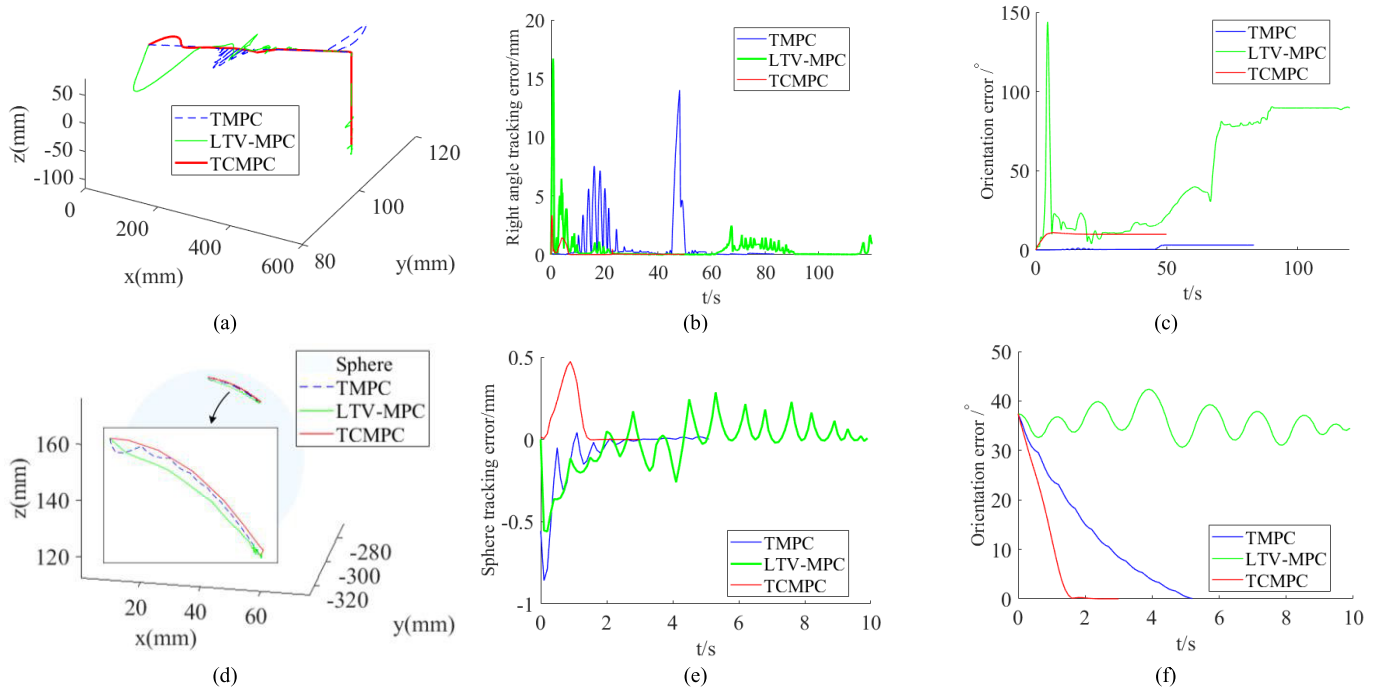


Fig. 9. Comparative simulation results of different MPCs. (a)–(c) Tracking performance, tracking error, and orientation error for a right-angle trajectory with the start point (45, 97, 67) and endpoint (500, 110, -100), respectively. (d)–(f) Spherical trajectory tracking with ${}^E F_s$ set to (0.3, -1.5, -2).

TABLE I
COMPARISON OF CONVERGENCE TIME AND TRACKING ERROR DATA FOR DIFFERENT MPC METHODS

Time / RMSE Model	Right-angle trajectory tracking			Spherical trajectory tracking		
	Converge time (t)	position (mm)	Orientation (deg)	Converge time (t)	position (mm)	Orientation (deg)
TMPC	83.2000	2.3391	2.0591	5.2000	0.2112	17.2855 (converge to 0)
LTV-MPC	119.9000	1.3992	60.1361	-	0.1903	35.8248 (disconverge)
TCMPC	49.9000	0.4287	9.8166	3.0000	0.1545	15.9092 (converge to 0)

$\tilde{t}_i = [t_i; \mathbf{u}_{i-1}]$ is the augmented state as described in [41], and α denotes the coefficient for the relaxation factor ε .

The setpoint segmentation interval of TCMPC and LTV-MPC is set to the same as λ in TCMPC. The explicit trajectory tracking method is also implemented in the LTV-MPC as a constraint. Comparative experiments are conducted on right-angle trajectories and spherical trajectory tracking as shown in Fig. 9(a)–(f).

3) *Simulation Analysis*: The simulation results shown in Figs. 6 and 7 indicate that with appropriately chosen weight parameters, the maximum root-mean-square error (RMSE) of position tracking does not exceed 1 mm, and the RMSE of end-effector orientation changes does not exceed 0.03 rad. For accurate positioning, the maximum RMSE does not exceed 0.3 mm. Analysis of variance (ANOVA) shows that the tracking error does not differ significantly under different force feedback conditions ($F = 1.45$ and $p = 0.2206$). It is also observed that these errors are most pronounced during the initial stage of the planning. The results demonstrate the excellent tracking accuracy and robustness of TCMPC. Moreover, the errors can be reduced by increasing the control horizon N or shortening the planning step size λ , as demonstrated in Figs. 6(e) and (f) and 7(e) and (f). The tracking accuracy or time cost can be maintained by adjusting the value of λ and N .

In the tracking control for cases 1-3 and 2-3 shown in Figs. 6(e) and (f) and 7(e) and (f), as described in Lemma 1 in Appendix A (see the Supplementary Material), increasing w_r and w_a improves the trajectory tracking performance. However, once they reach a certain value, the tracking performance remains basically unchanged. For example, the RMSE is 1.5837 mm for $w_r = 5e1$, 0.4303 mm for $w_r = 5e2$, and 0.4290 mm for $w_r = 5e3$. Similarly, the RMSE is 0.2981 mm for $w_a = 5e0$, 0.1934 mm for $w_r = 5e1$, and 0.1920 mm for $w_a = 5e3$.

The comparative results in Fig. 9 and Table I show that the proposed TCMPC outperforms other MPC methods in both tracking accuracy and convergence speed. Compared to other MPCs, the convergence speed and accuracy of TCMPC are, on average, improved by 42.48% and 44.08%, respectively. Since TMPC does not involve trajectory correction, its tracking performance extremely depends on the segmentation accuracy of the trajectory (interpolation error) (see Appendix B.1 in the Supplementary Material). LTV-MPC uses the same correction method as TCMPC but suffers from larger errors and slower performance due to its linear approximation. Moreover, dynamics based on the Cartesian state significantly degrade orientation tracking, even with high penalty coefficients.

From the manipulative technique tracking and force control results shown in Fig. 8, it is obvious that the proposed

TABLE II
RELEVANT EQUIPMENT AND MODEL SPECIFICATIONS FOR EXPERIMENT

Equipment	Specification
Host System	Ubuntu 18.04, ROS Melodic, PyQt Software
Controller	STM32
Force/Torque Sensor	Hypersen FT060 Series, 6-axis, ± 0.01 N·m (torque), ± 0.2 N (force)
Tension-Compression Sensor	SBT640-5kg by Spar, $\pm 0.25\%$ Full Scale (2kg)
Insert-Lift and Twisting Motors	IG28ET018 Model
Acupuncture Needle	0.30 mm $\times$ 30 mm (Stainless Steel, Tapered Type)
Human tissue model	Biomimetic Materials
Global Camera/ Hand-Eye Camera	Intel RealSense Depth Camera D415

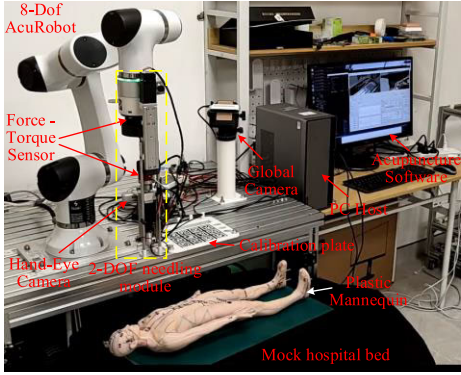


Fig. 10. Experimental system of the acupuncture robot.

controller effectively suppresses trajectories that exceed $F_{\max}$. It keeps the force tracking error within the threshold range, with velocity jitter primarily occurring during the initial stage of force changes. The size of threshold ν significantly impacts tracking performance. When ν is small, the force control task exhibits hysteresis, preventing timely switching to the safe velocity tracking task. This is related to the dynamic characteristics of the system (i.e., a low-frequency pole exists). If the value of ν is set to be small, it will take a long time for the system to enter such an error range. With a larger ν , switching between force control and velocity tracking becomes more prompt, although this results in increased jitter for both force and velocity. Additionally, the frequency of the technique also affects the switching between force control and velocity tracking. As shown in Fig. 8(b) and (e), the higher frequency of the trigonometric technique leads to less prompt switching compared to the lower frequency polynomial technique, given the same impedance filter and threshold settings. We also analyze the selection of control gains in (43)–(47). Please refer to Appendix F (see the Supplementary Material) for details.

B. Experiment

1) *System Setup*: The designed experimental system for the acupuncture robot is shown in Fig. 10. The equipment specification is detailed in Table II.

The global camera is fixed in position, while the hand-eye camera is mounted on the end of the acupuncture mechanism. Based on previous research by Zhang et al. [44], the extrinsic

calibration results are

$${}^0T_{G_c} = \begin{bmatrix} 0.611 & 0.522 & -0.594 & 65.647 \\ 0.791 & -0.403 & 0.459 & -540.990 \\ -0.001 & -0.751 & -0.6596 & 288.533 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (55)$$

$${}^E T_{L_c} = \begin{bmatrix} -0.027 & 0.998 & 0.044 & -31.017 \\ -0.999 & -0.027 & -0.003 & 41.469 \\ -0.002 & -0.044 & 0.999 & -147.073 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (56)$$

2) *Experimental Results*: They are given as follows.

a) *Trajectory tracking control experiment in rough positioning*: To verify the effectiveness and accuracy of the acupoint localization algorithm and to assess the acupuncture robot's ability to track polygonal trajectories, an acupoint is marked on a human model. The horizon N is set to 10, and the control frequency is set to 10 Hz. The tracking weight coefficient is set to $w_r = 5e2$, and the planning step size value is $\lambda = 1/4N$. The acupuncture point measurement algorithm based on circular markers has been previously discussed [45]. The experimental results are shown in Fig. 11.

The experimental results show that at the initial stage, the trajectory tracking control of the acupuncture robot is guided by the global camera. As the end-effector approaches the acupuncture point, the hand-eye camera progressively measures the precise position of the acupuncture point, allowing accurate control to reach the desired location. The polygonal trajectory effectively avoids potential collision risks between the robot and the operating table or human body, with the entire trajectory tracking error being less than 2 mm.

b) *Trajectory tracking control experiment based on bending perception*: The bending situation under different needle forces is shown in Appendix E.1 (see the Supplementary Material). It can be observed that the needle tip forces and bending conditions confirm to the beam deformation model depicted in Fig. 3(a).

To further validate the sphere trajectory tracking capability, two sets of needle bending conditions from Appendix E.1 (see the Supplementary Material) are selected for trajectory tracking experiments. Initially, ${}^E F_s = [0.105, -0.479, -5.419]^T$ and ${}^E F_s = [0.423, -0.054, -5.309]^T$ are perceived, and the sphere center coordinate is set to $P = [-283.621, -270.067, -12.752]^T$ with the needle length $r = 30$ mm. Set $N = 10$. The control frequency is set to 10 Hz. $w_a = 5$ and $\lambda = 1/3N$. The results are shown in Fig. 12. It can be observed that the maximum spherical tracking error is less than 0.3 mm. During the planning process, the angle between F_e and the z -axis may increase, but as the spherical planning proceeds, the angle gradually decreases and corrects the bending, with the largest needle's bending degree reduced by 61%.

It is worth noting that TCMPC and the adaptive impedance control are separated because they conflict with the control of acupuncture mechanism. More specifically, the manipulative

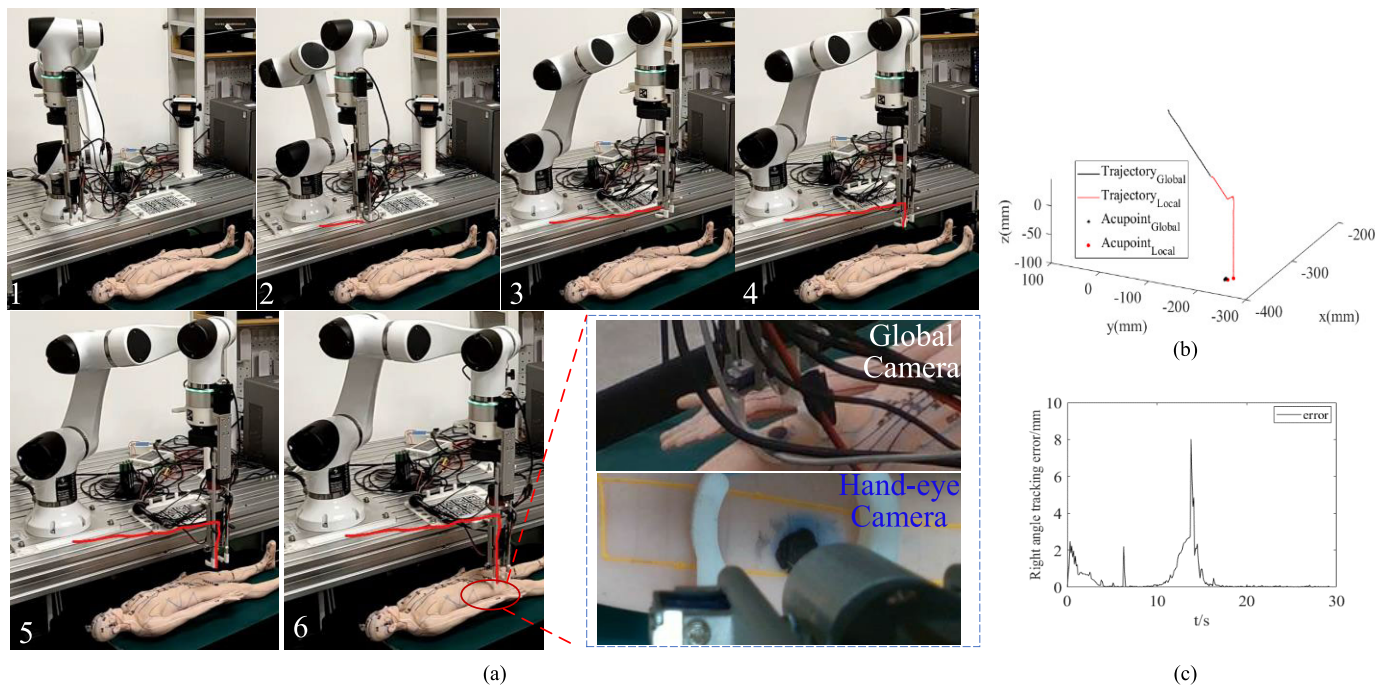


Fig. 11. Performance of the TCMPC by Appendix C.1 (see the Supplementary Material). (a) Snapshots of acupuncture robot. (b) Tracking trajectory and acupoint. (c) Trajectory tracking error.

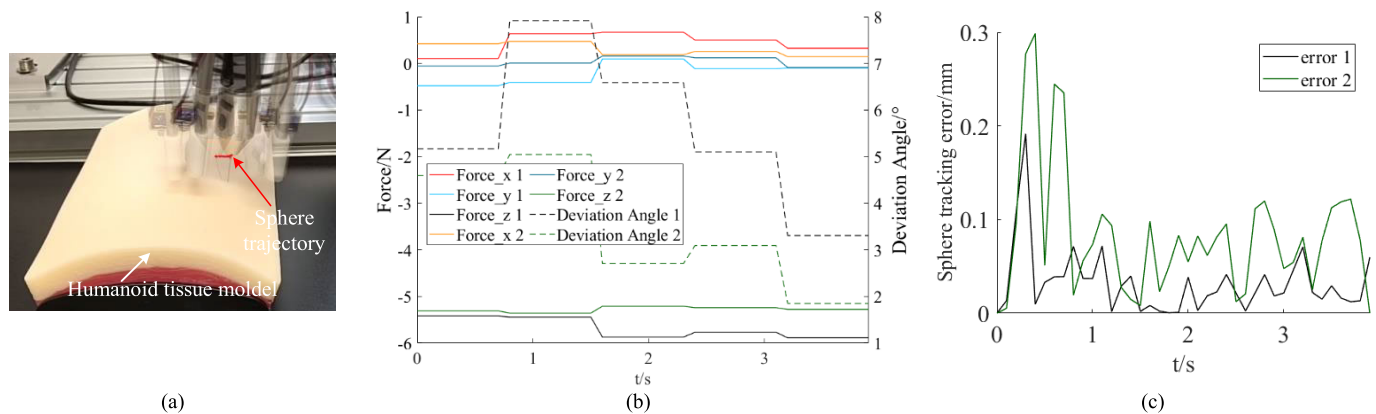


Fig. 12. Performance of TCMPC implemented by Appendix C.2 in the Supplementary Material (1 and 2 represent different force sensing feedback). (a) Bending correction. (b) Force variation. (c) Trajectory tracking error.

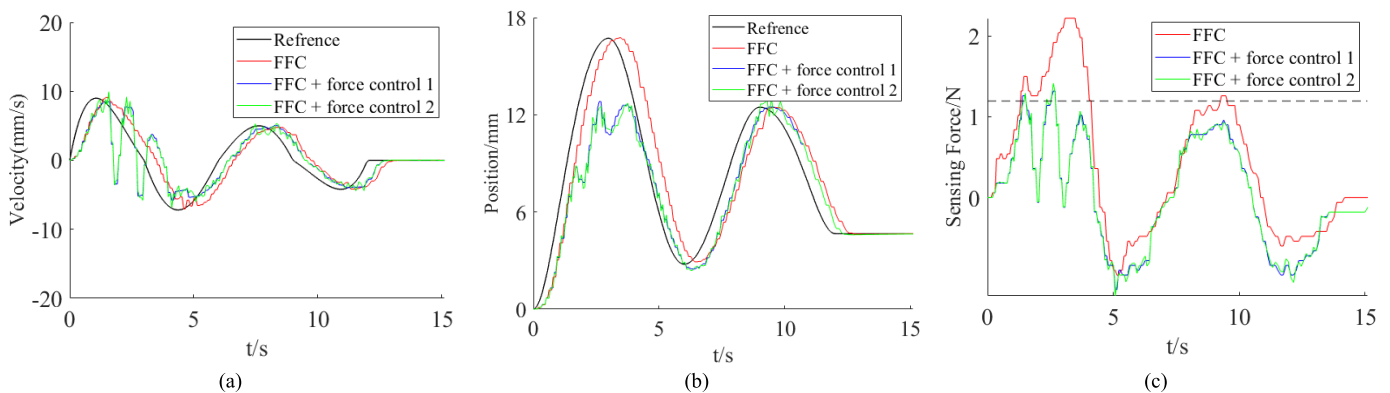


Fig. 13. Safety tracking performance of FFC + Adaptive FC (1 and 2 represent the trial numbers). (a) Velocity tracking. (b) Position tracking. (c) Force variation.

technique tracking is performed only after the bending of the needle is corrected.

c) Adaptive impedance control experiment for manipulative technique: To validate the effectiveness of FFC and

adaptive impedance control for acupuncture manipulative techniques tracking and force control, the technique is simulated according to the following polynomial function:

$$\dot{x}_d = \begin{cases} 0.0090 \cdot t(t-3)(t-5)(t-8)(t-10), & 0 < t < 3 \\ 0.0072 \cdot t(t-3)(t-5)(t-8)(t-10), & 3 < t < 6 \\ 0.0050 \cdot t(t-3)(t-5)(t-8)(t-10), & 6 < t < 9 \\ 0.0042 \cdot t(t-3)(t-5)(t-8)(t-10), & 9 < t < 12 \\ 0, & t > 12. \end{cases} \quad (57)$$

In the experiment, the FFC gain coefficient k_F is set to 1; the adaptive impedance parameters are set to $\gamma_1 = 5$, $\gamma_2 = 10$, $\gamma_3 = 0.5$, $\nu = 0.1$, and $F_{\max} = 1\text{N}$; and the controller's sampling frequency is 20 Hz. After multiple trials, the experimental results are shown in Fig. 13. The results indicate that the pure FFC closely matches the reference trajectory [Fig. 13(b)], with a maximum control error within 1 mm. The actual velocity curve can closely simulate the reference velocity curve. Without force control, the maximum force control error reaches 2.23 N. However, with adaptive impedance control, the experiment shows significant force control effectiveness, with a maximum force control error of 0.3 N. It effectively suppresses needle insertion trajectories that exceed $F_{\max}$ and achieves fast and accurate tracking for trajectories and speeds that do not exceed $F_{\max}$.

VII. CONCLUSION

In this article, an 8-DOF intelligent acupuncture robot system has been developed. A modeling and control method that integrates needle bending compensation with manipulative techniques was proposed. It allows for safe trajectory planning in operational space and achieves accurate, efficient, and robust trajectory tracking control through TCMPC. The tracking RMSEs for the right-angle trajectory and spherical trajectory are 3.1527 and 0.1103 mm, respectively. During the needling process, the force and bending of the needle were precepted and controlled to ensure the safety and accuracy of the contact operation. The largest needle's bending degree was reduced by 61%. An adaptive impedance control model considering the contact model has been established. This model can control force in various contact environments, detect operational forces in real time, and autonomously select between force control and manipulative technique tracking tasks. The maximum force control error is less than 0.3 N. Both simulation and experimental results confirm the effectiveness of the proposed methods.

The system adopts a marker-based visual perception and positioning scheme. For flexibility and dynamic clinical environments, markerless vision systems need further investigation. Additionally, the more precise nonlinear model of needle-tissue interaction should be considered. Designing the efficient nonlinear controller remains a key research challenge. Furthermore, exploring the adaptive update rate of the switching module threshold is another important direction for future work.

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The Appendix are available at https://github.com/RPCDLT-SYSU/AcuRobot/modeling_and_control/TC_appendix.pdf

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Weihao Li received the B.E. degree in intelligence science and technology from the School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen, China, in 2023, where he is currently pursuing the M.E. degree in control science and engineering.

His research interests include robotics and neural networks.



Junlong Tao received the B.E. degree in intelligent process equipment and control engineering from the School of Mechanical and Automotive Engineering, South China University of Technology, Guangzhou, China, in 2023. He is currently pursuing the M.E. degree in control science and engineering with the School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen, China.

His research interests include robotics and neural networks.



Yu Han received the Ph.D. degree from the Department of Automation, Shanghai Jiao Tong University, Shanghai, China, in 2011.

He was a Chief Engineer with the Robot Business Department, Jiangsu Automation Research Institute, Lianyungang, Jiangsu, China. He is currently a Professor with the School of Intelligent Systems Engineering, Sun Yat-sen University, Guangzhou, China. His research interests include intelligent manufacturing, robotics, and manufacturing automation of marine equipment.



Wanquan Liu (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from Shanghai Jiao Tong University, Shanghai, China, in 1993.

He once held the ARC Fellowship, the U2000 Fellowship, and the JSPS Fellowship and attracted research funds from different resources over \$2.4 million. He is currently the Dean of the School of Intelligent Systems Engineering, Sun Yat-sen University, Guangzhou, China. His current research interests include large-scale pattern recognition, signal processing, machine learning, and control systems.

Dr. Liu is on the editorial board for nine international journals.



Jianqing Peng (Member, IEEE) received the Ph.D. degree from the Department of Mechatronics and Automation, Harbin Institute of Technology, Shenzhen, China, in 2018.

He was a Post-Doctoral Research Fellow of Control Science and Engineering with Harbin Institute of Technology from 2018 to 2020. He was also a Research Associate (Honorary) with the Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Hong Kong, China, from 2019 to 2020. He is currently an

Associate Professor with the School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen. His research interests include robot large model technology, multimodal perception of robots, intelligent control of robots, and innovative applications of artificial intelligence technology in the field of robotics.